

Product Requirements Document AI Tool Design

About This Resource: This PRD documents the design and development of MiLLie, a GPT-based instructional coaching agent built to support educators working with multilingual learners. It covers the problem definition, target user personas, feature specifications, interaction flow, output quality criteria, and success metrics. It exists as both a design artifact and a development roadmap.

Problem-first design: The PRD opens with the problem before describing any solution. The problem statement grounds every subsequent design decision in a real and documented need — content-area teachers are making well-intentioned but ineffective instructional choices because they lack accessible, actionable guidance. Every feature in the tool traces back to closing that specific gap.

Interaction flow as a design constraint: The five-step UX flow is a constraint that governs every output. The agent is not permitted to generate materials before making recommendations, cannot skip the context-gathering step, and must maintain consistent output structure across all interactions. These rules exist because instructional coaching requires diagnosis before prescription.

Iterative build and measurement: The development plan is structured around four versioned releases with increasing quality targets, from 60-70% accurate generations at prototype stage to 80-90% at MVP release. Success is measured against a composite output rating rubric and an NPS target, making design decisions testable rather than subjective. This reflects a product thinking approach applied to an educational tool, treating pedagogy and usability as co-equal design requirements.

MiLLie Coach Agent: PRD

Purpose

Problem

Multilingual learners (MLs) are one of the fastest growing demographics of students in the U.S. and elsewhere. However, not all teachers receive adequate training or resources for them to confidently and effectively teach MLs, especially in core content area classes. In an effort to support MLs in accessing and engaging with core content, teachers often make well-meaning but ineffective instructional decisions, such as providing translations of all materials into students' home languages, or over-adapting materials or selecting texts that are oversimplified, resulting in the delivery of content that is below grade level. With an abundance of curriculum, supplemental, and AI tools, teachers have many options for what they can use, but not necessarily the know-how for how best to use them.

Solution

MiLLie the Multilingual Learning Coach is a GPT-based instructional coach that provides guidance for educators on best practices and follows through by executing on requests to adapt, scaffold, and design materials for multilingual learners using research-aligned strategies and contextualized examples.

Target User Personas

The MiLLie Coach is primarily intended to meet the needs of the following personas:

1. **First year and early career teachers:** These teachers are still getting accustomed to teaching in general and may have received little practical training for how best to support diverse groups of learners.
 - a. **Persona Needs:**
 - i. Actionable advice with simple “why it works” information for developing lessons and content
 - ii. Ready to implement scaffolds, adaptations, lesson ideas, and materials that save time while being pedagogically sound
2. **Teachers experiencing an influx of ML students:** Even experienced teachers require support and coaching when they are experiencing an influx of multilingual learners in their classes for the first time. The same may be true for a sudden influx of learners from different language backgrounds or mixed proficiency levels when previously the ML population was a more homogeneous group.
 - a. **Persona Needs:**
 - i. Actionable advice for adjusting existing lessons and content to accommodate a new population of learners
 - ii. Simple “why it works” information that shows the value of making the changes
 - iii. Ability to quickly generate modifications and materials to cut down on admin time

3. **Instructional coaches:** Instructional coaches, unless specifically trained ELL specialists, may not have the full background and practical knowledge on how best to support ML students, especially mixed language and proficiency groups.
 - a. **Persona Needs:**
 - i. Quick and easy to obtain actionable advice that can be passed along to teachers with a variety of backgrounds and needs
 - ii. Ability to generate modifications and materials as exemplars or for large amounts of diverse content and needs to support a cohort of educators

Features & Functionality

Core Features

1. **(Primary Function) Instructional Recommendation Engine** (i.e. how best to teach something)
 - a. **Input:** Goal, lesson, or challenge
 - b. **Output:** Two actionable recommendations or strategies
 - i. Summary of the request
 - ii. What to do
 - iii. Why it works
 - iv. What it targets
 - v. Examples
 - c. **Constraints:**
 - i. Must be actionable, not theory
 - ii. Must reflect best practices in ML teaching
 - iii. Each strategy must differ meaningfully from each other
2. **Guided Next Steps**
 - a. **1st Output:** Structured follow-up by closing out initial generation with an offer to further customize or generate based on more specific inputs
 - i. Asks for more information
 1. Ex: Tell me more about the specific lesson or topic you are working on so I can provide even more aligned strategies.
 - b. **Inputs:** Responses to requests for more details
 - i. Specifics about the lesson or topic
 - ii. Upload of materials
 - iii. More details about student demographics and/or needs
 - c. **2nd Output:** Generated more detailed/targeted advice or strategies
 - i. What to do
 - ii. Why it works
 - iii. Very specific examples
3. **On-Demand Material Modification or Generation**
 - a. **1st Output:** Offer next steps follow up after presenting actionable strategies (both generation 1 and 2, including offers to
 - i. Turn advice into a lesson plan
 - ii. Adapt uploaded content
 - iii. Create scaffolds
 - iv. Identify key grammatical or vocabulary structures
 - b. **Inputs:** Request by user

- i. Uploaded document
 - ii. Confirmation to move forward one of the offers
 - iii. Request to do something else
 - c. **2nd Output:** Generate requested materials in alignment with the selected strategy, proficiency level, grade-level, and other contextual inputs provided by user
- 4. Contextual Interpretation**
 - a. **Requests and Interprets:**
 - i. Proficiency level information
 - ii. Content area
 - iii. Grade level
 - iv. Task type
 - b. Infers when possible
 - c. Asked for direct input when needed
- 5. Instructional Reasoning Transparency**
 - a. **Output:** Explains why a strategy is effective, what language domains it addresses, what skills it develops
 - i. Aligned with best practices and research-backed approaches
 - ii. Builds trust and educates users

Interaction Flow (UX)

The tool must maintain a consistent user experience across all users and generations.

Step 1: User asks for help

- “How do I teach X?”
- “My students don’t understand this text.”
- “How do I incorporate speaking activities into my math class?”

Step 2: Agent responds with recommendations

- 2 targeted strategies
- Clear, structured, and consistent

Step 3: Agent prompts next step

- Requests more information and offers to tailor responses further
- Offers to build something from one of the strategies

Step 4: User selects direction

- Inputs more details (goes back to step 2)
- Asks another question (goes back to step 2)
- Makes a request for materials

Step 5: Agent generates aligned materials

Output Specifications & Quality Criteria

- 1. Recommendation Quality:**
 - a. Advice and recommendations are actionable, not generic theory
 - b. Each strategy is distinct (not the same thing reworded)
 - c. Grounded in multilingual learner pedagogy

- d. Aligned with common language proficiency frameworks
- e. Simple and friendly output that feels natural
- 2. Rules for Agent Responses & Questions:**
 - a. No proficiency level, grade level, or subject area provided → ask for it
 - b. Vague or unclear request → ask for clarification
 - c. Missing content when requested to scaffold, adapt, or build something → ask for it
 - d. Never ask for specific student PII
- 3. Output Structure:**
 - a. Summarize request (confirms it is answering the right question)
 - b. Strategy 1 + name
 - i. What to do
 - ii. Examples
 - iii. Why it works
 - iv. How it focuses on language
 - c. Strategy 2 + name
 - i. What to do
 - ii. Examples
 - iii. Why it works
 - iv. How it focuses on language
 - d. Offer to tailor further
 - e. Offer to generate materials
- 4. Guardrails:**
 - a. Avoid generating content before making recommendations
 - b. Avoid over-simplifying content so that it is no longer appropriate for the grade level or proficiency band
 - c. Avoid generic teaching advice like “try differentiating your instruction”
 - d. Avoid directly copying and regurgitating from knowledge base or online resources

Requirements & Steps to Build

Agent Requirements

- 1. GPT developer and model (ChatGPT)**
- 2. Clear, multi-shot prompt**
- 3. Comprehensive background knowledge that does not violate fair use or copyright:**
 - a. Best practices aligned with research-backed pedagogical models
 - b. Common proficiency level frameworks (reworded, not uploaded directly from WIDA/CEFR)
 - c. Sample lesson materials
 - d. Sample supplemental resource materials (e.g., graphic organizers, vocabulary resources)
 - e. Example questions and responses
 - f. List of common industry tools and applications with their purposes (e.g., Google Docs, Padlet, Kahoot, etc.)

High-Level Build Steps

1. Develop V1 prompt
2. Create V1 prototype agent (limited background knowledge)
3. Test and assess what background knowledge is required
 - a. Design output rating scale
4. Build collection of background knowledge materials
5. Revise V2 prompt and add background knowledge to create V2 agent
6. Test and assess output
 - a. Test with a variety of good and bad prompts
7. Iterate and create V3 agent with built-in feedback loop
 - a. Design embedded feedback tool (survey)
8. Test internally
 - a. Test with a variety of good and bad prompts
9. Test externally and gather feedback
 - a. Preferably test with a couple of actual ML teachers
10. Iterate and create MVP agent
11. Release with built-in feedback loop

Goals & Success Measures

Goal <i>Develop an AI-enabled ML educator coaching agent that...</i>	KPI	Measured By
Provides accurate and actionable recommendations and strategies for supporting multilingual learners that align with research-backed pedagogical models.	V2 (prototype) goal: 60-70% accurate/usable generations. V3 (test MVP) goal: 70-80% accurate/usable generation as rated by external users.	Output rating rubric (composite score; recommendation score, generation score) User feedback (survey)
Generates accurate and useable proficiency- and grade-level appropriate scaffolds, modifications, lesson plans, and/or materials that align with recommended strategies and best practices in multilingual learning.	V4 (release MVP) goal: 80-90% accurate/usable generations as rated by external users.	
Is useful enough that educators would use it in their classroom.	V3 (test MVP) goal: NPS between +10 and +30 V4 (early release MVP) goal: NPS between +30 and +50	User survey
Saves educators time when modifying or creating materials.	V3 (test MVP) goal: $\geq 40\%$ of users believe the tool would save time during their weekly planning and report. No target time saved; use to gather information.	User survey

	V4 (early release MVP): $\geq 50\%$ of users say the tool saves at least 1-2 hours of time per week.	
Helps educators better understand multilingual pedagogy and develop strategies they can use on their own.	V3 (test MVP) goal: $\geq 30\%$ of users say they found the advice informative. V4 (early release MVP): $\geq 50\%$ of users say they found the tool informative. <i>This goal is less defined and is more difficult to measure due to subjectivity and user background.</i>	User survey

Development & Release Plan

Project Tracker found here.

Future Steps

Based on the success of the ChatGPT based agent, the tool may be expanded into a standalone product using an OpenAI API. This requires far more development and will cost money for the various platforms and capabilities required to build and host the tool.